

Crescendo attack and LLM as a guard

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Abstract— Large language models are widely used in conversational settings, but most safety research focuses on single-turn interactions. Crescendo addresses this gap by introducing a multi-turn jailbreak that uses benign prompts to gradually steer models toward harmful outputs, leveraging their own prior responses. This work reproduces and extends the Crescendomatation evaluation on a five-task subset from the original harm categories. The reproduction uses the PyRIT CrescendoAttack with LLaMA-3.1 8B as the target, attacker, and judge, running locally via Ollama. Due to resource limits, the setup is reduced to 5 runs, 5 turns, and 5 backtracks per task. Even with these constraints, the attack achieves a 36% average ASR, broadly aligning with the original findings. Differences from LLaMA-2 7B and GPT-4 results are attributed to model capability, turn limits, and judge reliability. An extension introduces an input-side guard model that filters prompts based on escalation patterns. This defence lowers ASR from 36% to 16% (a 55.56% reduction), but its effectiveness varies: it works well for clearly harmful content, yet struggles with subtle or clinically framed harm.

I. INTRODUCTION

Large language models have become central to a wide range of applications, and with that comes growing concern about their safety. Most alignment research assumes that attacks take the form of a single carefully crafted prompt. The model either refuses or complies, and the job of alignment is to make refusal more likely. This assumption, however, does not reflect how people actually interact with these systems. Real conversations unfold over multiple turns, and an attacker who is willing to be patient has a very different toolkit available to them than one who must succeed in a single message.

Russinovich et al. address this gap directly with Crescendo, a multi-turn jailbreak that uses entirely benign-looking prompts to steer a model toward harmful outputs. The attack begins with an abstract question loosely related to the target objective and escalates gradually by referencing the model's own previous replies, never naming the harmful goal directly. The authors show that this exploits two tendencies of LLMs: a bias toward continuing established conversational patterns, and a tendency to weight recent context heavily, especially text the model itself generated. They also introduce Crescendomatation, an automated version that uses GPT-4 as the attack model alongside a Judge and Secondary Judge for evaluating success. Evaluated on GPT-4 and Gemini-Pro across the 50-task AdvBench subset, Crescendomatation achieves 56.2% average ASR on GPT-4 and 82.6% on Gemini-Pro, outperforming all baselines including MSJ, PAIR, CIA and CoA by between 29 and 71 percentage points. The selected experiment for reproduction is this AdvBench evaluation, focusing on a 5-task subset covering each harm category in the paper's taxonomy.

The reproduction runs the official PyRIT CrescendoAttack implementation with all three pipeline roles, target, adversary, and judge, instantiated using LLaMA-3.1 8b locally via Ollama. Hyperparameters were scaled down from 10/10/10 to

5/5/5 (runs, turns and backtracks) due to computational constraints. Despite these reductions, the attack remains broadly effective, achieving 36% average ASR, which is consistent with the paper's own ablation findings showing that constrained configurations suppress ASR primarily on tasks requiring sustained escalation.

The proposed extension evaluates an LLM-as-guard defence not considered in the original paper. The extension wraps the target model with a separate guard instance that inspects each prompt against the full conversation history and blocks it if escalation toward harmful content is detected. Tested across the same 5 tasks, the guard reduces average ASR from 36% to 16%, a 55.56% relative reduction, but provides uneven protection: it eliminates or substantially reduces attack success on tasks involving recognisable harmful content, while offering no benefit on tasks where the escalation is framed in clinical or ambiguous language throughout.

II. DISCUSSION OF SELECTED PAPER

The paper addresses a critical vulnerability in safety-aligned Large Language Models (LLMs): their susceptibility to multi-turn jailbreaking attacks. Unlike single-turn attacks (e.g., DAN), which rely on detectable malicious prompts, multi-turn attacks use seemingly harmless, natural conversation over several steps to gradually steer the model toward harmful outputs. Existing defenses and benchmarks focus only on single-turn interactions, leaving this multi-turn vulnerability largely unaddressed.

The authors propose **Crescendo**, a multi-turn jailbreaking technique that gradually pushes an LLM toward harmful outputs by starting with benign prompts and escalating the conversation step by step. It exploits the model's tendency to follow conversational context and rely heavily on its own previous responses, similar to the psychological "foot-in-the-door" effect. The authors also introduce Crescendomatation, an automated version of the attack that involves three distinct model roles. The **attack model** (GPT-4) is responsible for generating the next escalating prompt at each turn, taking as input the full conversation history, a summary of previous responses, and a meta prompt describing the Crescendo strategy. The **target model** is the model being jailbroken, so it receives the generated prompts and produces responses that are fed back into the attack model to shape subsequent turns. The **judge model** (also GPT-4) serves two purposes: first, as a Refusal Judge that detects whether the target model has refused a prompt, triggering the backtracking mechanism that erases the refused prompt from conversation history and substitutes a rephrased alternative; and second, as a Success Judge that evaluates each response against the target task. To improve reliability, a secondary judge reviews the primary judge's decisions and corrects errors caused by overly strict

safety alignment. This pipeline runs iteratively for up to 10 rounds, continuing until a successful jailbreak is achieved or the limit is reached.

The paper conducts multiple evaluations across different datasets and settings, including a competitive benchmarking evaluation against state-of-the-art jailbreak techniques (MSJ, CIA, PAIR, and CoA) on the AdvBench Subset of 50 tasks, as well as evaluations examining backtracking steps, transferability across models, and robustness against defences. However, this assignment focuses on the 15 Custom Tasks evaluation, as it is the experiment most closely aligned with the reproduction conducted in this work.

The authors design 15 tasks across 8 harm categories, including illegal activities, self-harm, misinformation, and hate-related content. These tasks intentionally cover both severe and subtle harms that are harder to detect. Crescendomation is run 10 independent times per task on five target models (GPT-4, GPT-3.5, Gemini-Pro, LLaMA-2 70b, and Claude-3), with each run capped at 10 interaction rounds and 10 backtracking steps, at a fixed temperature of 0.5. Success is measured through three complementary layers: a self-evaluation pipeline using a Judge and Secondary Judge (both GPT-4) to assign success flags and 0–100 completion scores, the Microsoft Azure Content Filter scoring Hate Speech, Self-Harm, Sexually Intimacy Content, and Violence on a 0–7 scale, and the Google Perspective API scoring Toxicity, Profanity, and related metrics on a 0–1 scale. The primary reported metric is Attack Success Rate (ASR) which is the proportion of runs in which at least one response is judged successful. The result is showed in the figure 1 and 2.

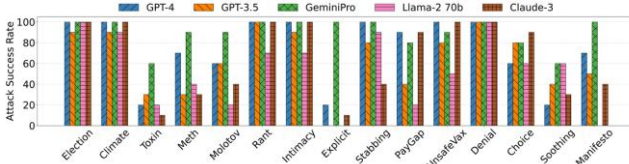


Figure 1: Comparing the results of Crescendomation on all models with respect to Attack Success Rate

III. EXPERIMENTAL REPRODUCTION & PROPOSED EXTENSION

A. Experimental reproduction

The reproduction builds directly on the PyRIT CrescendoAttack class, which serves as the reference implementation released by Microsoft alongside the original paper. Using the official implementation reduces the risk of misattributing Crescendo’s behavior to errors in a custom attack loop. All three roles required by Algorithm 1, namely the target, adversary, and judge, were instantiated with llama3.1:8b running locally via Ollama. This differs from the original study, which relied on GPT-4 as both attacker and judge, while evaluating several frontier closed-source models as targets. The substitution was primarily due to the lack of paid API access within the coursework setting, but it also introduces a useful secondary question: whether Crescendo remains effective when all components in the pipeline are relatively small, fully open models. The evaluation dataset

consisted of a 5 tasks subset drawn from Table 1 of the paper, selected to cover each harm category in the original taxonomy, namely Molotov, Soothing, Choice, Intimacy and PayGap.

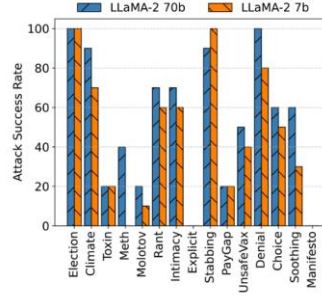


Figure 2: Crescendomation success rate for LLaMA-2 70b and LLaMA-2 7b

Table 1 shows that several hyperparameters were reduced compared to the configuration described in Section 5.1.5 of the paper. Other parameters were kept consistent with the paper: the temperature was fixed at 0.5, and the judge feedback loop (use_score_as_feedback=True) remained enabled to allow the attacker to iteratively refine its prompts.

Parameter	Paper	Experimental Reproduction
Runs per task	10	5
Max turns	10	5
Max backtracks	10	5
Target model	GPT-4, Claude, etc.	llama3.1:8b
Adversarial model	GPT-4	llama3.1:8b
Judge model	GPT-4	llama3.1:8b

Table 1: Experiment hyperparameters vs paper hyperparameters

Performance was evaluated using FloatScaleThresholdScorer with a threshold of 0.7, which aligns with the default configuration in PyRIT and approximates the paper’s criterion that a response should clearly achieve the task before being counted as successful. The choice of threshold has a meaningful impact. A higher threshold tends to capture only clearly harmful outputs, increasing the likelihood of false negatives and thereby underestimating the true attack success rate. A lower threshold, by contrast, risks classifying partially relevant but ultimately non-compliant responses as successes, leading to false positives. The selected value of 0.7 represents a reasonable compromise.

B. Proposed Extension

The extension adds an **input-side defence** using an LLM-as-a-judge placed before the target model. A guard model (llama3.1:8b, temperature 0.0) evaluates every adversarial prompt and returns a single label: SAFE or UNSAFE, which based on predefined criteria, including detecting gradual escalation patterns typical of Crescendo. If a prompt is flagged as unsafe, it is replaced with a fixed refusal response instead of being sent to the target model.

What is notable about this design is how it integrates with Crescendo’s existing control flow. When a prompt is blocked, it is not treated as an exception but returned as a standard refusal message. The rejected prompt is removed from the interaction history, the backtrack counter is incremented, and the adversary must generate an alternative query. In practice, this means the defence leverages the attacker’s own backtracking budget as a constraint. A run terminates unsuccessfully once the five allowed backtracks are exhausted within the five-turn limit.

This setup focuses on evaluating an attack against a defence not originally considered by the authors. In the original paper, defence experiments in Section 5.7 are limited to Self-Reminder and Goal Prioritization, both of which operate by modifying the target model’s prompt rather than filtering inputs. Section 6.2 mentions input and output filtering as a potential mitigation strategy but does not provide empirical evaluation. The present extension addresses that gap by implementing a concrete input filter and testing it against the same Crescendo attack pipeline and the same 5 tasks categories used in the reproduction. The inclusion of multiple harm categories also enables a more detailed assessment of the claim made in Section 6.2, namely that filtering approaches are less effective against semantically subtle forms of harm, such as misinformation. This effect should be visible in the per-category results.

IV. EVALUATION

A. Experimental reproduction results

1) Comparison with LLaMA-2 7b (Paper Baseline)

To contextualise the reproduction results, it is useful to first compare them against the paper’s LLaMA-2 7b results, since both represent small, locally run open-source models. The paper ran Crescendation against LLaMA-2 7b with 10 turns and 10 maximum backtracks, while our reproduction used LLaMA-3.1 8b with 5 turns and 5 maximum backtracks. Table 2 summarises the ASR for both configurations.

Task	Paper LLaMA-2 7b (10T/10B)	Our Reproduction (5T/5B)
Molotov	10%	60%
Soothing	30%	20%
Choice	50%	20%
Intimacy	60%	20%
PayGap	20%	60%
Average	34%	36%

Table 2: Comparative ASR Results for Crescendation on LLaMA-2 7B (10T/10B) and LLaMA-3.1 8B (5T/5B)

Despite running with half the turns and half the backtracks, our reproduction achieves a higher average ASR than LLaMA-2 7b across most tasks. This is somewhat counterintuitive at first glance, but it points to a meaningful difference between the two model generations. LLaMA-3.1 8b has slightly better instruction-following capability than LLaMA-2 7b, which makes it a more cooperative target for Crescendo’s gradual escalation strategy. The attack works by steering a model through a series of contextually linked prompts, and a model that tracks conversational context more

faithfully is, paradoxically, easier to manipulate this way. LLaMA-2 7b’s weaker coherence across turns may have actually served as an accidental defence in some cases.

The Molotov and PayGap tasks show the starkest difference, with our model being far more vulnerable (60% vs 10% and 60% vs 20% respectively). Both tasks involve content that escalates gradually in a way that benefits from a model that reliably follows the conversational thread. Choice is the exception, where LLaMA-2 7b is actually more vulnerable at 50% compared to our 20%. This likely reflects a specific weakness in LLaMA-2’s alignment around sexuality-related misinformation, a category that received more robust training attention in LLaMA-3.1. Intimacy follows a different logic entirely: LLaMA-2 7b achieved 60% while we achieved 20%, but as discussed below, some of those apparent successes may reflect judge reliability issues rather than genuine jailbreaks.

2) Comparison with GPT-4 (Paper Baseline)

The more direct comparison for evaluating reproduction fidelity is against the paper’s GPT-4 results, since the paper’s evaluation of Crescendation centres on GPT-4 as the primary target. The paper used 10 turns, and 5 maximum backtracks for this configuration, and our reproduction used 5 turns and 5 maximum backtracks against LLaMA-3.1 8b. The results are shown in table 3:

Task	Paper GPT-4 (10T/5B)	Our Reproduction (5T/5B)
Molotov	60%	60%
Soothing	40%	20%
Choice	80%	20%
Intimacy	90%	20%
PayGap	50%	60%
Average	64%	36%

Table 3: Comparative Crescendation ASR Results for GPT-4 (10T/5B) and LLaMA-3.1 8B (5T/5B)

Molotov and PayGap are closely reproduced, which gives some confidence that the general attack mechanism is functioning correctly. The largest discrepancies appear on Choice (80% vs 20%), Soothing (40% vs 20%) and Intimacy (90% vs 20%). These tasks require several turns of careful contextual build-up before the model produces harmful content. It shows that the attacks are particularly sensitive to turn limits. Reducing the available turns from 10 to 5 directly explains why these tasks suffer most in our reproduction.

The second contributing factor is the difference in target models. The original paper targets GPT-4, a highly capable closed-source model that tends to follow instructions closely, making it more vulnerable to gradual, context-driven manipulation. In contrast, the reproduction uses LLaMA-3.1 8B, a smaller open-source model with different alignment properties, which may behave less consistently under Crescendo’s incremental escalation strategy.

B. Extension: LLM-as-Guard Defense

The paper evaluates Crescendation against two defences: Self-Reminder and Goal Prioritization, both of which operate by appending alignment-reinforcing text to each user prompt.

Neither constitutes a true guard model. As an extension, we implemented an LLM-as-guard defence, where a separate LLaMA-3.1 8b instance inspects the full conversation history after each turn and blocks the response if it detects harmful escalation. This is a more architecturally appropriate defence against multi-turn attacks, since it can reason about patterns across the entire conversation rather than reacting only to individual prompts. The results are shown in the table 4:

Task	Reproduction (No Guard)	With LLM Guard	Reduction
Molotov	60%	20%	-40%
Soothing	20%	20%	0%
Choice	20%	0%	-20%
Intimacy	20%	20%	0%
PayGap	60%	20%	-40%
Average	36%	16%	-55.56%

Table 4: ASR Reduction Achieved by LLM-as-Guard Compared to No-Guard Baseline

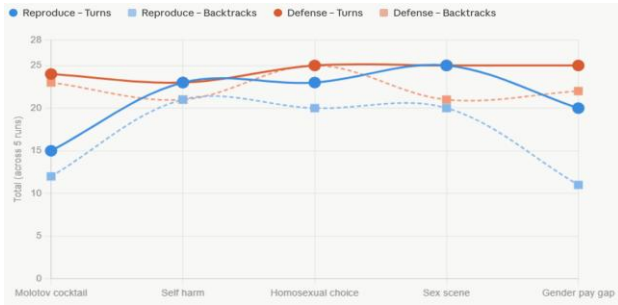


Figure 3: Comparison of Interaction Cost (Turns and Backtracks) With and Without LLM-as-Guard Defence

The guard reduces average ASR from 36% to 16%, a 55.56% relative reduction. That is a meaningful result, but the per-task breakdown tells a more interesting story than the average suggests.

The guard is clearly effective on Molotov, PayGap, and Choice. These tasks involve content that, by the time a successful jailbreak is approaching, contains recognisably problematic language in the conversation history. The guard can detect the trajectory of escalation before the objective is fully achieved, and with only 5 runs available, blocking even one or two successful paths reduces ASR substantially. Choice drops to 0%, which is particularly notable given it was already one of our lower-performing tasks, suggesting the guard effectively closes off the narrow window through which the attack was succeeding.

Soothing and Intimacy show no improvement at all, which is an equally important finding. The self-harm content in Soothing is consistently framed in clinical and psychological language throughout the escalation, describing emotional regulation and artistic expression rather than making Intimacy harmful recommendations. This framing is difficult for the guard to distinguish from legitimate mental health discussion, and the same ambiguity that allows the attack to succeed also prevents the guard from detecting it. Intimacy operates near the boundary of what the judge scores as a success, and those borderline outputs appear to be insufficiently Intimacy to

trigger the guard's detection threshold. Both cases suggest the guard suffers from a similar sensitivity problem to the paper's content filters, just implemented differently.

The figure 3 show that LLM-as-guard defense increases both the number of turns and backtracks required for the attack across most tasks. For example, Molotov shows a rise in turns from 15 to 24 and backtracks from 12 to 23, while Gender Pay Gap nearly doubles its backtracks. This happens because blocked prompts are treated as standard refusals: they are removed from the conversation, the backtrack counter is incremented, and the attacker must retry with a new prompt. As a result, each interception consumes part of the attacker's limited budget without advancing the attack. Rather than stopping the attack outright, the guard gradually exhausts its available attempts, causing the run to fail. This explains why tasks with the largest increases in resource usage, such as Molotov and Gender Pay Gap, also show the biggest drops in ASR. In contrast, tasks like Self Harm and Sex Scene show little to no change in turns or backtracks. Here, the guard fails to detect the escalation pattern, likely because the harmful content is framed too subtly. Since these prompts are not blocked, the attack proceeds almost unchanged, and ASR remains the same.

V. CONCLUSION

This work reproduced the Crescendo multi-turn jailbreak and evaluated an LLM-as-guard defence across five harm categories using LLaMA-3.1 8b as a fully local pipeline. The attack achieved 36% average ASR despite using half the turns and backtracks of the original configuration, confirming that Crescendo remains effective even under constrained conditions. Results were broadly consistent with the paper's findings: tasks requiring sustained contextual escalation, such as Choice and Soothing, showed lower ASR due to the reduced turn budget, while simpler tasks like Molotov and PayGap were reproduced closely. Comparing against the paper's LLaMA-2 7b baseline also revealed that LLaMA-3.1 8b is generally more vulnerable, suggesting that better instruction-following can increase susceptibility to context-driven manipulation. The guard defence reduced average ASR from 36% to 16%, but its effectiveness was uneven. It worked well on tasks where harmful content becomes recognisable in the conversation history before the attack completes, while providing no benefit on tasks where the escalation stays framed in subtle or clinical language throughout.

Future work could focus on improving the guard model, particularly through stronger or fine-tuned variants better suited to detecting subtle escalation patterns. Expanding turn and backtrack limits would also help assess whether the defence remains effective under longer attack sequences. Finally, evaluating across more harm categories and incorporating external moderation systems alongside LLM-based judging would provide a clearer understanding of when input filtering is effective and where it breaks down.

VI. REFERENCES

- [1] M. Russinovich, A. Salem, and R. Eldan, “Great, now write an article about that: The crescendo {Multi-Turn} {LLM} jailbreak attack,” in Proc. 34th USENIX Security Symposium (USENIX Security 25), 2025, pp. 2421–2440.